

sikit Validation Guide

Hardware shopping list, test board options, and a step-by-step procedure for verifying sikit's predictions against real measurements.

sikit's math is anchored by unit tests and a published-reference-data suite (IPC-2141A canonical cases, Polar Si9000 values, monotonicity sanity). To take validation one tier up — comparing predictions against a physical PCB — you need a test board with known geometry and an instrument that can measure S-parameters or impedance. This guide tells you exactly what to buy and how to use it.

1. Test Equipment

Sikit's validation does not require a benchtop \$20k VNA. A USB-connected “NanoVNA”-class instrument is sufficient for everything FR-4 boards can practically carry (up to ~6 GHz). All recommended VNAs export Touchstone (.s2p) files that sikit's `File > Open Touchstone (Ctrl+T)` action loads directly.

USB VNAs

Model	Range	Price (USD)	Recommendation
NanoVNA H4	50 kHz – 1.5 GHz	\$60 – \$80	Entry level; USB 2.0 / sub-GHz validation
NanoVNA-F V2 (SAA-2)	50 kHz – 3 GHz	\$120 – \$180	Good for FR-4 hobbyist work
LiteVNA 64	50 kHz – 6.3 GHz	\$150 – \$200	Recommended sweet spot
PocketVNA	100 kHz – 4 GHz	\$250 – \$500	Cleaner build, PC-tethered
Copper Mountain S5180	300 kHz – 18 GHz	\$5k – \$15k	Lab grade

Why the LiteVNA 64. The 6.3 GHz upper limit covers PCIe Gen 2, USB 3.0 (5 Gbps), the upper edge of FR-4 useful bandwidth, and the full sikit `synthesize_channel` sweep we generate by default (10 MHz to 20 GHz, but FR-4 loss dominates the result above 6 GHz anyway).

Accessories

Item	Why	Price (USD)
SMA cables, phase-stable 30 cm	Connect VNA to test board	\$15 – \$30 ea
Edge-launch SMA female × 8	Solder onto trace endpoints	\$3 – \$5 ea
50 Ω chip terminators, 0402 × 8	Single-port S11 if cable count limited	\$0.20 ea
VNA calibration kit (OSL)	Calibrate before measuring	Usually included with VNA
NanoVNA-Saver (PC app)	Touchstone export from NanoVNA family	Free

Minimum viable setup cost

Item	Cost
LiteVNA 64	\$170
2× SMA cables, 30 cm	\$30
8× edge-launch SMA female	\$20
Custom 4-layer test board, 5 pieces (JLPCB)	\$80

Item	Cost
Shipping (boards + parts)	\$30
Total	≈ \$330

Below this, the zero-hardware fallback is sikit's reference-data unit tests (already shipped) plus comparing against scikit-rf's bundled .s2p files in Python. Cost: \$0. Validates the math but not the physical engine.

2. Test Board Options

Three credible paths to a board you can measure against sikit's predictions, ranked from least-effort to most-effort.

Option A — Buy a pre-characterized compliance board

Source	Cost (USD)	What you get
PCIe SIG Gen 3 compliance load board	\$200 – \$600 (eBay used)	4-port S-params, eye masks, calibrated reference
Tektronix BERTScope fixtures	\$300 – \$1500 (used)	Channel S-params, multiple impedance targets
TI / ADI SerDes evaluation boards	\$50 – \$300 (new)	Documented diff pairs, SMA breakouts, datasheet S2P
Mini-Circuits / Pasternack RF reference fixtures	\$150 – \$400	Calibrated channels, less SI-specific

Pro: zero design work, ships pre-characterized with vendor-supplied .s2p files. Con: optimized for someone else's problem (specific protocol, specific stack-up); you pay \$400+ to validate one configuration.

Option B — Custom test coupon (recommended)

Design a board with multiple traces of known geometry, send to JLCPCB with **impedance-controlled fabrication enabled**, receive boards plus a TDR scan report from the fabricator. The TDR report is your ground truth for the closed-form and FDM engines, with no measurement effort on your side.

Suggested traces on the coupon:

Trace	Geometry (on JLC 4-layer 1.6 mm stackup)	sikit predicts
50 Ω microstrip	W = 0.36 mm, F.Cu, prepreg 0.21 mm	$\approx 50 \Omega$
75 Ω microstrip	W = 0.17 mm, F.Cu	$\approx 75 \Omega$
100 Ω diff microstrip	W = 0.15 mm, S = 0.15 mm, F.Cu	$\approx 100 \Omega$ diff
90 Ω diff microstrip	W = 0.10 mm, S = 0.20 mm, F.Cu	$\approx 90 \Omega$ diff (USB-style)
50 Ω stripline	W = 0.20 mm on In1.Cu, B = 0.4 mm	$\approx 50 \Omega$
TDR baseline	Straight 50 Ω , no discontinuities	Calibration reference

Each trace ~75 mm long, terminated at both ends with edge-launch SMA pads. Total board ~100 × 80 mm, fits comfortably under JLCPCB's smallest pricing tier.

Cost: ~\$30 base + \$25 impedance-control surcharge + \$25 shipping $\approx$ **\$80 for 5 boards**. Lead time 7–14 days.

Option C — Vendor evaluation boards with documentation

Common SI-grade EVBs that ship with documented trace dimensions or vendor-supplied .s2p files. Less precise than Option B because the EVB is built for a specific chip demo, but useful if a vendor's eval is already on your bench.

Vendor / Family	Why it's useful for sikit	Approx cost
TI TUSB1310A USB 3 PHY EVM	Documented 90 Ω diff traces	\$50 – \$150
TI LMK / DS-series SerDes evals	Published S2P channels	\$100 – \$300
Analog Devices ADRV9xxx	High-speed I/Q channels with .s2p	\$200 – \$500
NXP LayerScape RDB	10G Ethernet channels with measurements	\$100 – \$300

3. Step-by-Step Validation Procedure

Assumes you chose Option B (custom test coupon from JLCPCB) and the LiteVNA 64.

Step 1 — Design and order the board

1. Open the sikit-testcoupon KiCad project (separate repo to be set up).
2. Verify the stackup matches JLC04161H-3313: F.Cu (35 μm) / prepreg (0.21 mm, $\epsilon_r = 4.5$) / In1.Cu (17.5 μm) / core (1.06 mm, $\epsilon_r = 4.6$) / In2.Cu (17.5 μm) / prepreg (0.21 mm) / B.Cu (35 μm).
3. File > Plot > Manufacturing Outputs > Gerber + drill files. Use KiCad's JLCPCB defaults preset.
4. Zip the gerber output directory.
5. Upload to jlcpcb.com / Instant Quote. Drop the zip.
6. Pick options: 4 layers, 1.6 mm, ENIG surface, Impedance Control = enabled (this is the key one — adds \$25–40 but unlocks the TDR scan report), Stackup = JLC04161H-3313.
7. Place the order: ~\$80 total. Lead time 7–14 days.

Step 2 — Assemble

1. Solder edge-launch SMA female connectors onto the prepared pads at each trace endpoint.
2. Inspect every joint under magnification (10 $\times$). A reflowed or partially-wetted joint adds uncalibratable parasitics and ruins the measurement.
3. If a trace has only one SMA (single-port), populate a 0402 50 Ω chip terminator on the far-end footprint.

Step 3 — Calibrate the VNA

1. Connect LiteVNA to PC via USB. Open NanoVNA-Saver.
2. Select sweep range (e.g. 1 MHz to 3 GHz, 401 points).
3. Run OSL calibration with the supplied standards: Open at port end, Short at port end, Load at port end. NanoVNA-Saver guides you through each step.
4. For 2-port S-parameter measurements, also run a Through calibration with both cables connected to each other.
5. Save the calibration to a named slot so you can reuse it.

Step 4 — Measure each trace

1. Connect cable Port 1 to one SMA end of the trace.
2. Connect cable Port 2 to the other SMA end (for 2-port S-params) or terminate the far end with 50 Ω (for 1-port S11 / TDR).
3. Run the sweep.
4. Export to Touchstone (.s2p) via NanoVNA-Saver > File > Save / Export. Name the file after the trace, e.g. *trace_50ohm_microstrip.s2p*.
5. Repeat for every trace on the board.

Step 5 — Compare against sikit

1. Launch sikit. Open the same KiCad project (File > Open KiCad PCB or Ctrl+O).

2. Trace impedance overlay: hit Ctrl+1 (50 Ω target), Ctrl+2 (90 Ω), Ctrl+3 (100 Ω). Each trace's color shows sikit's predicted error band vs target.
3. Cross-check with JLC's TDR report: each controlled-impedance trace has a measured Z_{meas} in that PDF. Compare to sikit's prediction. Expected agreement: $\pm 10\%$ closed-form, $\pm 15\%$ FDM.
4. Open each measured .s2p: File > Open Touchstone for eye (Ctrl+T). Sikit applies a PRBS-7 NRZ TX and renders the eye diagram with USB 2.0 HS mask overlay.
5. Compare to sikit's own synthesized prediction: Analyze > Eye diagram — synthesized (Ctrl+Y), pick the same net. The two eyes should overlay closely once you account for the connector + cable loss in the measured version.
6. Optional: enable the FDM solver (Analyze > Use FDM solver for impedance) and repeat the overlay. The FDM result is slower but uses the real stackup parsed from KiCad rather than closed-form defaults.

Acceptance criteria

Engine	Expected agreement vs measured	Reason for residual error
Closed-form (IPC-2141A)	$\pm 10\%$ on Z_{meas} for 50 / 75 / 100 Ω cases	Formula model vs Hammerstad / fab tolerances
FDM (in-house 2-D Laplace)	$\pm 15\%$ on Z_{meas} at default mesh $h = W / 25$	Mesh discretization + edge fattening
Synthesized channel S21	Within 1–2 dB up to 3 GHz, more above	Lossless TEM + scalar $\tan \delta$ vs real dispersion
Eye diagram height / width	Within 10% of measured numerical values	TX ramp model is single-pole RC, not IBIS-AMI

Persistent disagreement larger than these bands probably indicates one of: wrong stackup (double-check JLC's stackup PDF for your exact panel), a soldering issue on the SMA (re-inspect, re-flow), or VNA calibration drift (re-run OSL).

Appendix — Useful links

What	Where
JLCPCB capabilities + stackup	jlcpcb.com/capabilities
JLC04161H-3313 stackup PDF	jlcpcb.com / Help & Support / Production Capabilities
NanoVNA-Saver (Windows / Linux / Mac)	github.com/NanoVNA-Saver/nanovna-saver
LiteVNA 64 firmware + docs	github.com/Cho45/LiteVNA-firmware
scikit-rf test .s2p files	github.com/scikit-rf/scikit-rf/tree/master/skrf/data
PCIe SIG compliance program	pcsig.com / Compliance Program
IPC-2141A standard	ipc.org / Standards / IPC-2141A
Wadell, Transmission Line Design Handbook	Artech House, ISBN 0-89006-436-9
sikit GitHub	github.com/UnsignedChad/sikit